INCH-POUND

MIL-C-85049/1A
30 November 1990
SUPERSEDING
MIL-C-85049/1
16 January 1980
MS3057E
10 May 1982

MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, BACKSHELL, ENVIRONMENTAL, CABLE SEALING, STRAIGHT, GROUNDING (WITHOUT STRAIN RELIEF), CATEGORY 1C (FOR MIL-C-5015 SOLDER TYPE, V THREAD OF MS310X CLASSES A, B, C OR K CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-85049.

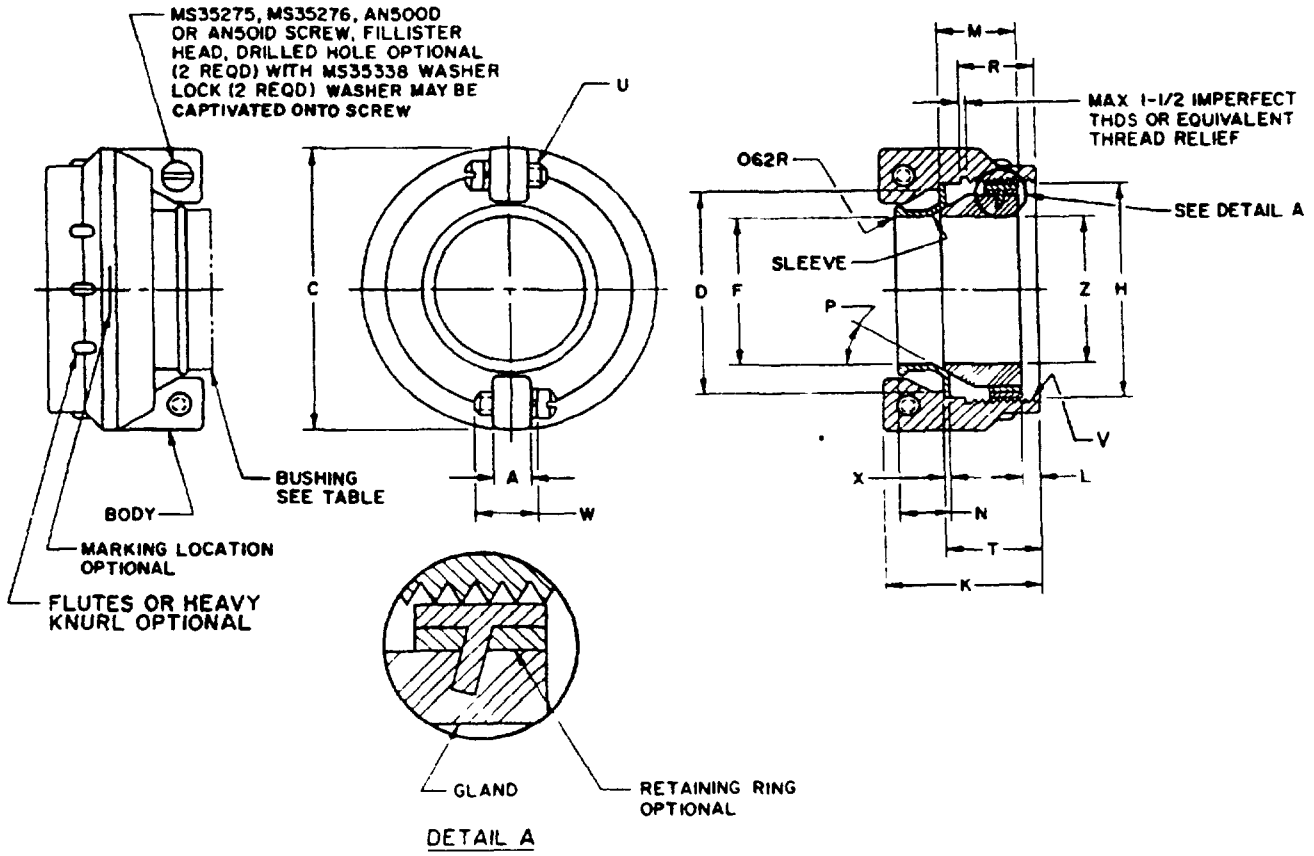


FIGURE 1. Configuration and dimensions.

AMSC N/A
FSC 5935

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TABLE I. Dimensions.

Dash No.	Connector Shell Size (Ref) 1/	A +.031 -.016	C Max	D +.000 -.020 Dia Gland	F +.020 -.000 Dia	H +.000 -.020 Dia	K Max	L Min	M ±.031	N	P ±2°	R Min	T	U Thread	MS35338 Washer, Lock Dash Number
-3B	8S and 10S	.188	.812	.389	.231	.441	1.031	.156	.473	.254	20°	.531	.641	6-32UNC-2A	136
-4B	10SL, 12S and 12		.937	.510	.315	.560									136
-6B	14S and 14		1.062	.627	.440	.676					30°				136
-8B	16S and 16		1.188	.744	.515	.800									136
-10B	18		1.312	.870	.614	.926	1.094		.537	.281			.672		136
-12B	20 and 22	.250	1.562	1.037	.738	1.107	1.219		.625	.320			.719	8-32UNC-2A	137
-16B	24 and 28		1.750	1.286	.926	1.357			.662	.344					137
-20B	32	.312	2.250	1.590	1.200	1.670	1.344			.406	35°	.562	.797	10-32UNF-2A	138
-24B	36		2.375	1.844	1.363	1.920	1.547		.940			.781	1.016		138
-28B	40		2.625	2.083	1.611	2.162									138
-32B	44	.375	2.812	2.332	1.865	2.412	1.734	.188	1.047	.500			1.172		138
-40B	48		3.312	2.832	2.365	2.912	1.781						1.234		138

Dash No	Fillister Head Screw				V Thread	W Min	X	Z Dia Approx (Gland)		Bushing (Ref)	Max Cable Dia
	AN500D Dash Number	MS35275 Dash Number	AN501D Dash Number	MS35276 Dash Number				Free	Closed		
-3B	6-5	227			.500-28 UNEF-2B	.281	.025 +.010 -.003	.219	.027	MS3420-3A	.209
-4B	6-5	227			.625-24 UNEF-2B			.312	.094	MS3420-4A	.302
-6B	6-5	227			.750-20 UNEF-2B			.438	.230	MS3420-6A	.428
-8B	6-5	227			.875-20 UNEF-2B			.531	.312	MS3420-8A	.515
-10B	6-5	227			1.000-20 UNEF-2B		.032 +.012 -.003	.625	.375	MS3420-10A	.614
-12B	8-6	243			1.188-18 UNEF-2B	.344		.750	.445	MS3420-12A	.738
-16B	8-6	243			1.438-18 UNEF-2B			.938	.594	MS3420-16A	.926
-20B			10-7	262	1.750-18 UNS-2B	.406	.051 ±.006	1.250	.922	MS3420-20A	1.200
-24B			10-7	262	2.000-18 UNS-2B			1.375	.922	MS3420-24A	1.363
-28B			10-7	262	2.250-16 UN-2B			1.625	1.180	MS3420-28A	1.611
-32B			10-7	262	2.500-16 UN-2B		.064 ±.006	1.875	1.375	MS3420-32A	1.865
-40B			10-7	262	3.000-16 UN-2B			2.375	1.875	MS3420-40A	2.365

1/ Clamp mates with connector shell size listed in the column.

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INCH	mm	INCH	mm	INCH	mm
0.003	0.08	0.531	13.49	1.219	30.96
0.006	0.15	0.537	13.64	1.234	31.34
0.010	0.25	0.560	14.22	1.250	31.75
0.012	0.30	0.562	14.27	1.286	32.66
0.013	0.33	0.594	15.09	1.312	33.32
0.016	0.41	0.614	15.60	1.344	34.14
0.020	0.51	0.625	15.88	1.357	34.47
0.025	0.64	0.627	15.93	1.363	34.62
0.027	0.68	0.641	16.28	1.375	34.92
0.032	0.81	0.662	16.81	1.438	36.53
0.051	1.30	0.672	17.07	1.547	39.29
0.062	1.57	0.676	17.17	1.562	39.67
0.064	1.63	0.719	18.26	1.590	40.39
0.094	2.39	0.738	18.75	1.611	40.92
0.156	3.96	0.744	18.90	1.625	41.28
0.188	4.78	0.750	19.05	1.670	42.42
0.209	5.30	0.781	19.84	1.734	44.04
0.219	5.56	0.797	20.24	1.750	44.45
0.230	5.84	0.800	20.32	1.781	45.24
0.231	5.87	0.812	20.62	1.844	46.84
0.254	6.45	0.870	22.10	1.865	47.37
0.281	7.14	0.875	22.23	1.875	47.63
0.302	7.67	0.922	23.41	1.920	48.77
0.312	7.92	0.926	23.52	2.000	50.8
0.315	8.00	0.937	23.80	2.083	52.91
0.320	8.13	0.940	23.88	2.162	54.91
0.344	8.74	0.969	24.61	2.250	57.15
0.375	9.53	1.000	25.4	2.332	59.23
0.389	9.88	1.016	25.81	2.365	60.07
0.406	10.31	1.031	26.19	2.375	60.33
0.428	10.87	1.037	26.34	2.412	61.26
0.438	11.13	1.047	26.59	2.500	63.5
0.440	11.18	1.062	26.97	2.625	66.68
0.441	11.20	1.094	27.79	2.812	71.42
0.445	11.30	1.107	28.12	2.832	71.93
0.473	12.01	1.172	29.77	2.912	73.96
0.510	12.95	1.180	29.97	3.000	76.2
0.515	13.08	1.200	30.48	3.312	84.12

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REQUIREMENTS:

1. Design and construction:
   Dimensions and configuration: See figure 1 and table I.

2. Lubricant, VV-P-236, shall be applied to threads and surfaces of sleeve.

3. Optional retaining ring shall be bonded to gland.

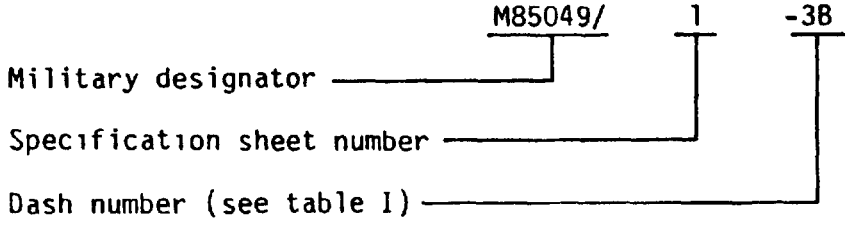
4. For sizes 3 through 10, serrations shall be covered with rubber. For sizes larger than 10, serrations may be covered with rubber. Depth of rubber shall not exceed .013.

5. Material and finish: See table II.

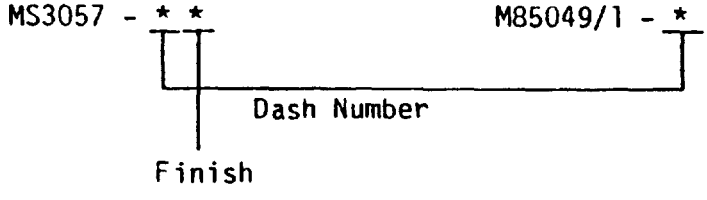
6. Qualification: See MIL-C-85049, Category 1C. Information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Avionics Center, Attention: Code B/444, 6000 East 21st Street, Indianapolis, IN 46219-2189.

7. Fluid immersion requirements not applicable.

8. Part number: See example below:



9. Supersession data:



Old	New
A	W
C	W
N	W

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TABLE II. Material and finish.

Material		Finish
Body	Aluminum alloy in accordance with MIL-C-85049.	1/ W
Screws and washers	Corrosion Resistant steel in accordance with MIL-C-85049.	Passivate - per QQ-P-35
Sleeve	Aluminum alloy in accordance with MIL-C-85049.	In accordance with MIL-C-5541. The portion of the sleeve in contact with the body shall be coated with polytetrafluoroethylene resin for sizes 3 through 10, optional.
Gland	Rubber AMS 3209, durameter A, 70 ± 5	None
Retaining ring optional	Brass SAE 70, hard temper only	None

1/ W finish is not for use in space applications.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are based upon 1.00 inch = 25.4 mm and shall be in accordance with ASTM E380, paragraph 4.5, Method A and ASTM E29, rounding-off method.
3. Unless otherwise specified, tolerances shall be as follows: Decimals ±.016, angles ±2°.
4. Dimensions apply after plating.
5. Dimensions and tolerancing in accordance with ANSI Y14.5.
6. Remove all burrs and sharp edges.
7. For additional closedown of the Z dimension, bushings MS3420-3A through MS3420-40A may be used. Diameter Z shall be free of obstructions.
8. Bushings (MS3420) are not supplied with the backshell and must be procured separately.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force - 11, 99
DLA - ES

Preparing activity:
Navy - AS

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